SHOVON MANDAL

Lecturer, Department of Computer Science and Engineering | Northern University of Business and Technology Khulna, Bangladesh
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RESEARCH PROFILE

Computer Science and Engineering lecturer and graduate researcher working on secure distributed systems and applied intelligent computing. My research includes blockchain-based digital payments, NFC and IoT transaction security, IOTA-based decentralized communication, embedded systems security, applied machine learning, natural language processing, and explainable AI. I have authored peer-reviewed conference publications in distributed systems, cybersecurity, machine learning, NLP, and AI-enabled applications, with additional manuscripts currently under review. I am interested in pursuing Ph.D. or thesis-based M.S. opportunities aligned with these research areas, with research assistantship or teaching assistantship support.

RESEARCH INTERESTS

Secure Distributed Systems; Blockchain-Based Digital Payments; IoT and NFC Transaction Security; Applied Cryptography; Embedded Systems Security; Applied Machine Learning; Natural Language Processing; Explainable AI.

EDUCATION

M.Sc. Eng. in Computer Science and Engineering **Jul 2022 - Present**

Khulna University of Engineering and Technology, Bangladesh

Status: Only thesis defense remaining | **Thesis:** Blockchain Technology Based Smart E-Payment Strategy for Secure Cashless Payment

Supervisor: Dr. Pintu Chandra Shill | **Research Area:** Secure Transaction Verification in Distributed Networks (Blockchain, NFC)

Relevant Coursework: Principles of Cryptography; Advanced Network Security; Cloud Computing; Neural Networks.

B.Sc. in Computer Science and Engineering **Spring 2018 - Spring 2022**

BRAC University, Bangladesh

CGPA: 3.80/4.00 | **Academic Standing:** Highest Distinction

Thesis: Blockchain-based Micropayment System for Secured Cashless Small Payments

Supervisor: Md. Golam Rabiul Alam, PhD | **Research Area:** Applied Cryptography for Secure Distributed Systems

PUBLICATIONS AND RESEARCH

Published Conference Papers

1. **S. Mandal** and P. C. Shill, "Ensuring Security and Efficiency in Digital Payments using Blockchain with NFC," *2024 International Conference on Knowledge Engineering and Communication Systems (ICKECS)*, Chikkaballapur, India, 2024, pp. 1-7, DOI.
2. A. Ghosh, **S. Mandal**, M. J. Islam, K. M. Islam, S. Dhar and R. Baidya, "Comparative Evaluation of Multiple CNN Architectures for Dermoscopic Skin Lesion Classification using ISIC Dataset," *2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)*, Chittagong, Bangladesh, 2026, pp. 1-6, DOI.
3. M. Keya, **S. Mandal**, S. Dhar, H. M. Tahsin, et al., “Q-learning based Automated Message Multicast in Gossip Protocol for Node Confirmation in IOTA Tangle,” in *2024 27th International Conference on Computer and Information Technology (ICCIT)*, 2024, pp. 387-392, DOI.

4. A. Ghosh, F. T. Pinki, R. Baidya, F. Alam and **S. Mandal**, "A Machine Learning and DEM-Based Approach for Predicting Sea Level Rise and Assessing Coastal Risk in Bangladesh Using AI Insights," *2026 IEEE 15th International Conference on Communication Systems and Network Technologies (CSNT)*, Al-Khobar, Saudi Arabia, 2026, pp. 199-204, DOI.

Accepted / In Press Conference Papers

1. **S. Mandal**, et al., "Revolutionizing Academic Learning and Research with the Integration of Leap Motion and ARCore in Android," in *16th International Conference on Computing, Communication and Networking Technologies (ICCCNT)*, Indore, India, 2025, in press.
2. A. Ghosh, **S. Mandal**, et al., "Advancing Sentiment Analysis Through Natural Language Processing and Deep Learning," in *16th International Conference on Computing, Communication and Networking Technologies (ICCCNT)*, Indore, India, 2025, in press.
3. A. Ghosh, A. Datta, **S. Mandal**, P. M. Swadesh, K. A. Ahmed, and M. H. Molla, "Beyond Single Transformers: A BanglaBERT-XLM-RoBERTa Stacking Framework for Code-Mixed Emotion Recognition," accepted for presentation at the *3rd IEEE International Conference on Computing, Applications and Systems (COMPAS)*, Dhaka, Bangladesh, 2026.
4. M. H. Islam, N. A. S. Jiku, **S. Mandal**, P. M. Swadesh, and A. Ghosh, "Automated Visual Inspection of Tilapia Fish Quality Using CNN and Transfer Learning," accepted for presentation at the *2026 IEEE International Conference on Optics, Machine Learning and Emerging Technology (OMLET)*, 2026.
5. D. Biswas, M. Hasan, A. Ghosh, **S. Mandal,** and K. A. Ahmed, "Explainable Hybrid CNN-BiLSTM-Transformer Network Intrusion Detection with Unknown-Attack Rejection," accepted for presentation at the *12th IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering (WIECON-ECE 2026),* 2026.

SELECTED RESEARCH AND DEVELOPMENT PROJECTS

Blockchain and NFC-Based Secure Cashless Payment Prototype | *Thesis-Supporting Research Implementation*

- Developed a blockchain- and NFC-based secure payment prototype with Arduino hardware experimentation.
- Implemented Arduino and Python modules for transaction processing, device-level verification, and embedded system testing.
- **Repository:** github.com/shovon-mandal/NFC | **Related Output:** ICKECS 2024 paper on Blockchain with NFC.

IOTA-Based Decentralized Trust and Transaction Verification System | *Web-Based Research Prototype*

- Developed a web-based prototype to demonstrate IOTA-supported payment interaction and decentralized transaction flow.
- Implemented front-end and backend components for payment demonstration, user interaction, and experimental validation.
- **Repository:** github.com/shovon-mandal/IOTA-Website | **Related Output:** ICCIT 2024 paper on IOTA Tangle.

ACADEMIC APPOINTMENTS AND TEACHING EXPERIENCE

Lecturer, Department of CSE | *Northern University of Business and Technology Khulna, Bangladesh* **Mar 2024 - Present**

- Teach undergraduate CSE courses and laboratories in computer architecture, digital logic, OOP, technical writing, and optimization.
- Prepare course materials, assessments, lab tasks, and support student mentoring and research engagement.

Courses Taught: Computer Architecture; Digital Logic Design; Object-Oriented Programming with Java; Technical Writing; Linear Programming and Combinatorial Optimization; Digital Logic Design Lab; Linear Programming and Combinatorial Optimization Lab

Lecturer (Adjunct), Department of CSE | *Imperial College of Engineering, Khulna, Bangladesh* **Jul 2022 - Jun 2024**

- Delivered undergraduate lectures, laboratory sessions, assessments, and academic support for core CSE courses.

Courses Taught: Database Management Systems; Computer Peripherals and Interfacing; Computer Architecture and Organization; Computer Graphics; Web Engineering; Microprocessor and Microcontrollers

Teaching Assistant, Department of CSE | *Khulna University of Engineering and Technology, Bangladesh* **Jul 2023 - Dec 2023**

• Assisted faculty members in laboratory instruction, assignment evaluation, academic record preparation, and student support.

Courses Assisted: Embedded Systems; Electronic Devices and Circuits

Student Tutor, Department of CSE | *BRAC University, Bangladesh* **Feb 2022 - Apr 2022**

• Provided individual and group academic support to undergraduate students through tutoring sessions and guided academic assistance.

Courses Assisted: Electronic Devices and Circuits

ADDITIONAL TECHNICAL PROJECTS

Student Management Android Application | *Course Project* | GitHub Repository

• Developed an Android-based student management application using Android Studio.

Embedded Hardware Prototypes | *Course and Hardware Projects*

• Built traffic-light and SIM-enabled fire-alarm prototypes using microcontroller-based interfacing.

TECHNICAL SKILLS

Programming: Python, Java, C, MATLAB, MIPS Assembly, JavaScript

Web and Database: MySQL, PHP, Laravel, HTML, CSS, JavaScript

Research or Data Tools: LaTeX, SPSS, Microsoft Excel, Google Sheets, Git, API integration, workflow automation

Systems and Hardware: Arduino prototyping, NFC payment modules, embedded systems, hardware interfacing, digital electronics

Blockchain and Security: Blockchain security, applied cryptography, secure distributed systems, IOTA-based trust mechanisms

Languages: English, Bengali (Native)

HONORS, SCHOLARSHIPS, AND ACADEMIC SERVICE

- Highest Distinction, BRAC University.
- Vice Chancellor's List, BRAC University - 5 consecutive semesters.
- Dean's List, BRAC University - 2 consecutive semesters.
- Merit-based Scholarship, BRAC University.
- Judge, Project Showcasing - CSE Fest 2025
- Executive Member, BRAC University Computer Club.
- Executive Member, Robotics Club of BRAC University.
- Executive Member, BRACU Mongol Tori, Next Generation Mars Rover Team 4.0.

References and supporting documents are available upon request.